Koa Shen

Mechanical Engineer

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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Dynamics, Mechanics of Materials, Circuits, MATLAB. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Built a Python/Drake simulation to check a 30-DOF, 305-body beamline assembly for interference; millisecond collision checks exposed at least five physical pinch points and 10+ actionable design changes.
- Cross-checked CoACD convex-hull collision results with a secondary mesh triangle-triangle distance query, and compared model geometry with the physical assembly to catch outdated CAD.
- Performed statics and dynamics hand calculations to qualify stages and mounts against crash and drop load cases for $100k+ detectors.
- Developed 15 reusable motion-stage models now being adopted across three LCLS assemblies; released Solid Edge designs with GD&T through Teamcenter PDM.

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- Verified precision-machined suspension features with go/no-go tests across 33 parts, holding clevis and rod-end bores within a 0.0004-in total tolerance window and bearing interfaces within 0.0005 in.
- Tested a redesigned steering wheel for ergonomics, strength, and FSAE compliance; FEA-guided iterations reduced mass from 5 lb to 1.7 lb.
- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation; the car passed FSAE technical inspection for the team's first time and competed in every dynamic event.
- Placed 2nd among UC teams and best among UC teams in endurance at 2026 Michigan FSAE Electric.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Diagnosed and repaired failures across 100+ public exhibits under continuous, unpredictable visitor use; led three major overhauls.
- Traced acrylic enclosure cracking to over-constraining fasteners and replaced them with a load-spreading bezel, avoiding $3,000+ in replacement costs; redesigned sand-damaged encoder mounts for serviceability.
- Documented CAD, CAM files, and remanufacturing instructions in Confluence for future repairs.

PROJECTS

Closed-Loop Cycloidal Actuator | Personal Project — Summer 2026 - Present

- Built PlatformIO C++ firmware for TMC2209 stepper control with real-time CSV telemetry logging time, step position, and measured angle.
- Bench-validated closed-loop position correction and missed-step detection with a 12-bit AS5600 encoder on a single-joint test setup.
- Developed a backlash test that preloads the shaft against a limit switch and uses encoder feedback to measure travel before the switch changes state.

FSAE Chassis Welding Fixtures | Gaucho Racing — Winter 2024 - Spring 2026

- Designed fixtures that located 81 chassis tubes within 0.050 in of CAD nodes; cut assembly time 50% for GR25 and a further 30% for GR26.

TECHNICAL SKILLS

Test and Validation: Bench testing, data logging, go/no-go inspection, fixture design, failure diagnosis, load-case analysis, technical inspection
Instrumentation and Software: Python, C/C++, MATLAB, PlatformIO, Arduino, AS5600 encoders, TMC2209 drivers, Git
Mechanical: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, FEA, CNC and manual machining, welding